

Decoupled Geometric Control of the Quadrotor UAV Based on INDI

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Abstract. This paper introduces a novel decoupled geometric control method for quadrotor UAVs, based on the principles of incremental nonlinear dynamic inversion (INDI). The proposed method leverages differential flatness and INDI to linearize the nonlinearity of the UAV system, allowing for effective control of both translational and rotational movements. By applying INDI, the controller achieves strong tracking capabilities, particularly for large-angle yaw movements, while ensuring asymptotic convergence of attitude and position tracking errors. To validate the effectiveness of the proposed controller, simulations comparing it with an undecoupled geometric controller are conducted. The results demonstrate that the decoupled geometric control algorithm offers consistent performance and precise trajectory tracking.

Keywords: Decoupled Geometric Control, Incremental Nonlinear Dynamic Inversion, Nonlinear Control

1 Introduction

Quadrotor unmanned aerial vehicles (UAVs) have recently garnered significant interest due to their vast potential, affordability, lightweight portability, ease of maintenance, and superior maneuverability. These UAVs have been successfully deployed in various applications such as target search, food delivery, and disaster rescue [1]. In scenarios such as navigating tight spaces or rapidly locating targets in earthquake-affected areas, there is a crucial need for quadrotors to exhibit responsive attitude control and high-speed flight. Accurate trajectory tracking becomes vital in complex environments; however, achieving this is challenging due to the strong coupling characteristics inherent in UAV attitude control [2]. Traditional flight controllers using Euler angles often face limitations due to potential singularity issues [3], prompting the development of quaternion-based controllers [4]. Nonetheless, these can experience unwinding events during actual flights, leading to discontinuous trajectories.

Numerous researchers have explored various methods to reduce the nonlinear effects on quadrotor UAV attitude control. Techniques employed include fuzzy controllers [5], neural network controllers [6], and sliding mode controllers [7],

among other algorithms. While these methods have mitigated the nonlinear impact on the UAV model, they have also led to a significant increase in computational burden [8], limiting their practical application.

Compared to other methods, Nonlinear Dynamic Inversion (NDI) is a practical nonlinear control algorithm. By accurately characterizing the system, NDI can generate linearized control laws and eliminate all nonlinearities [9, 10]. This method streamlines the control process, enhancing the stability and dependability of attitude control. However, NDI is highly sensitive to model mismatches [11]. Therefore, when utilizing NDI, attitude control performance and robustness can be improved by increasing modeling accuracy and calibrating the model appropriately.

To effectively handle the nonlinear dynamics and model uncertainty of the quadrotor UAV, this paper presents an improved geometric control law based on INDI [12]. In contrast to the approach in [13], which designs attitude controllers in $SO(3)$, the proposed controller achieves a decoupling of roll/pitch control and yaw control for the UAV. The roll/pitch controllers are designed based on the desired direction of total thrust, while the yaw controller is designed based on the desired direction of yaw motion. This decoupling architecture enables independent control of roll/pitch and yaw, enhancing the UAV's performance during agile flight.

The following are this paper's primary contributions: (a) The proposed INDI-based decoupled geometric controller is simpler to implement for agile flights than geometric controllers designed within $SO(3)$. (b) The quadrotor UAV's high-speed maneuvers induce unknown external disturbances, which are compensated for by the proposed INDI decoupled controller for yaw angle. The stability of this controller has been proven through Lyapunov stability analysis. (c) Simulation results verify that, compared to geometric controllers constructed in $SO(3)$, the proposed controller offers superior performance in terms of agile flight and system robustness.

2 The model of a quadrotor UAV

We define $\{\mathbf{I}\} = \{X^I, Y^I, Z^I\}$ as the inertial coordinate system and $\{\mathbf{B}\} = \{X^B, Y^B, Z^B\}$ as the body coordinate system in accordance with the right-hand rule principle. Define $\mathbf{e}_1 = [1, 0, 0]^T$, $\mathbf{e}_2 = [0, 1, 0]^T$, $\mathbf{e}_3 = [0, 0, 1]^T$. The quadrotor UAV's dynamic models are defined as follows :

$$\dot{\mathbf{p}} = \mathbf{v}. \quad (1)$$

$$m\dot{\mathbf{v}} = mg\mathbf{e}_3 - f\mathbf{R}\mathbf{e}_3 + \Delta_x. \quad (2)$$

$$\dot{\mathbf{R}} = \mathbf{R}\mathbf{S}(\mathbf{w}). \quad (3)$$

$$\mathbf{J}\dot{\mathbf{w}} + \mathbf{w} \times (\mathbf{J}\mathbf{w}) = \boldsymbol{\tau} + \Delta_R. \quad (4)$$

In the inertial coordinate system, the position vector is $\mathbf{p}(t) = (x, y, z)^T$, and its velocity vector is $\mathbf{v}(t) = (v_x, v_y, v_z)^T$. The UAV's mass and gravity acceleration are shown by the variables m and g , respectively, while the overall thrust produced by the rotor is indicated by f . The rotation matrix for the UAV from the body coordinate system to the inertial coordinate system is represented by the matrix $\mathbf{R} \in \mathbb{R}^{3 \times 3}$. In the body coordinate system, the angular velocity of the UAV is represented by the vector $\mathbf{w} = [w_1, w_2, w_3]^T$. The quadrotor UAV's rotational inertia characteristics are described by the matrix $\mathbf{J}$, which represents its inertia matrix. Unknown external disturbances are represented by the vectors Δ_x and $\Delta_R = [\Delta_1, \Delta_2, \Delta_3]^T$. To control the quadrotor UAV's attitude motion, the control input torque is represented by the vector $\boldsymbol{\tau} = [\tau_1, \tau_2, \tau_3]^T$. The definition of the function $\mathbf{S}(\mathbf{w})$ is as follows:

$$\mathbf{S}(\mathbf{w}) = \begin{bmatrix} 0 & -w_3 & w_2 \\ w_3 & 0 & -w_1 \\ -w_2 & w_1 & 0 \end{bmatrix}. \quad (5)$$

Rewrite (4) as:

$$J_1 \dot{w}_1 + (-J_2 + J_3)w_2w_3 = \tau_1 + \Delta_1. \quad (6)$$

$$J_2 \dot{w}_2 + (-J_3 + J_1)w_1w_3 = \tau_2 + \Delta_2. \quad (7)$$

$$J_3 \dot{w}_3 + (-J_1 + J_2)w_1w_2 = \tau_3 + \Delta_3. \quad (8)$$

The state of the UAV can be represented as the vector $[x, y, z, \psi]^T$, following the differential flatness characteristics described in [15]. The quadrotor UAV's position in the inertial frame is represented by the vector $\mathbf{p} = [x, y, z]^T$, with its yaw angle indicated by ψ . This paper aims to develop an INDI geometric control rule with yaw angle decoupling, enabling the quadrotor UAV to follow rapidly changing yaw trajectories even in the presence of unforeseen external disturbances.

3 Attitude Controller Design

Presuming that the quadrotor UAV's desired position trajectory in the inertial coordinate system is $\mathbf{p}_d = [x_d, y_d, z_d]^T \in \mathbb{R}^3$, and that the first body-fixed axis of the UAV's desired direction is $\mathbf{b}_{1d}(t) \in \mathbf{S}^2 = \{q \in \mathbb{R}^3 \mid \|q\| = 1\}$.

3.1 Roll/Pitch Attitude Control Subsystem

From (2), it is evident that the control resultant force of the UAV is $-f\mathbf{R}e_3 \in \mathbb{R}^3$. The resulting force is $\mathbf{R}e_3$ and it acts in that specific direction. Hence, to effectively regulate the UAV's orientation, it is vital to govern its condition. The goal is to construct an attitude control system that achieves the following conditions: as time approaches infinity, the matrix $\mathbf{A}$ approaches $-f\mathbf{R}e_3$. Additionally, we will examine the stability of the resulting coupled system, assuming a virtual control input $\mathbf{A}$.

Given the expression, the desired direction of the $\mathbf{b}_3$ axis is indicated as $\mathbf{b}_{3_d}$:

$$\mathbf{b}_{3_d} = -\frac{\mathbf{A}}{\|\mathbf{A}\|}. \quad (9)$$

Manipulating the orientation of the third axis $\mathbf{b}_3$ fixed to the body of the quadrotor UAV results in roll and pitch movements. In this part, we will develop a roll/pitch controller that utilizes (τ_1, τ_2) to manipulate the orientation of the third body-fixed axis $\mathbf{b}_3$ of the quadrotor UAV. The objective is for $\mathbf{b}_3$ to converge to $\mathbf{b}_{3_d}$ as t approaches infinity. The remaining component τ_3 will be utilised for the yaw control system.

Begin by formulating the equations of motion for the $\mathbf{b}_3$ axis in the body frame of the UAV on the two-dimensional spherical surface $\mathbf{S}^2$. Define $\mathbf{b}_i$ as the vector $\mathbf{R}\mathbf{e}_i$ in the unit sphere $\mathbf{S}^2$, where i belongs to the set $\{1, 2, 3\}$. Subsequently, it might be acquired:

$$\begin{aligned} \dot{\mathbf{b}}_3 &= \dot{\mathbf{R}}\mathbf{e}_3 = \mathbf{R}\mathbf{S}(w)\mathbf{e}_3 \\ &= \mathbf{R}\mathbf{S}(w)\mathbf{R}^{-1}\mathbf{R}\mathbf{e}_3 = \mathbf{R}\mathbf{S}(w)\mathbf{R}^T\mathbf{b}_3 \\ &= \mathbf{S}(\mathbf{R}w)\mathbf{b}_3 = \mathbf{w}_{12} \times \mathbf{b}_3. \end{aligned} \quad (10)$$

Since the cross product of $\mathbf{b}_1$ and $\mathbf{b}_2$ is zero, the value of $\mathbf{w}_{12}$ in $\mathbb{R}^3$ is:

$$\mathbf{w}_{12} = w_1\mathbf{b}_1 + w_2\mathbf{b}_2. \quad (11)$$

By employing the Taylor series expansion, we can derive the following expressions for (6) and (7):

$$\dot{w}_1 = \dot{w}_{10} + D_1(w, \Delta t) + \frac{\Delta\tau_1}{J_1} + \frac{\Delta_1}{J_1}. \quad (12)$$

$$\dot{w}_2 = \dot{w}_{20} + D_2(w, \Delta t) + \frac{\Delta\tau_2}{J_2} + \frac{\Delta_2}{J_2}. \quad (13)$$

$\Delta\tau_1 = \tau_1 - \tau_{10}$, $\Delta\tau_2 = \tau_2 - \tau_{20}$ is the incremental of $\boldsymbol{\tau}$. $D_1(w, \Delta t)$, $D_2(w, \Delta t)$ is defined as:

$$\begin{aligned} D_1(w, \Delta t) &= \frac{J_2 - J_3}{J_1}(w_2 + w_3)(w_2w_3 - w_{20}w_{30}). \\ D_2(w, \Delta t) &= \frac{J_3 - J_1}{J_2}(w_2 + w_3)(w_2w_3 - w_{20}w_{30}). \end{aligned} \quad (14)$$

Hence, the expression for $\dot{w}_{12}$ is:

$$\begin{aligned} \dot{w}_{12} &= (\dot{w}_{10} + D_1(w, \Delta t) + \frac{\Delta\tau_1}{J_1} + \frac{\Delta_1}{J_1})\mathbf{b}_1 \\ &\quad + (\dot{w}_{20} + D_2(w, \Delta t) + \frac{\Delta\tau_2}{J_2} + \frac{\Delta_2}{J_2})\mathbf{b}_2. \end{aligned} \quad (15)$$

The desired angular velocity $\mathbf{w}_{12}$ for the quadrotor UAV can be expressed as :

$$\dot{\mathbf{b}}_{3d} = -\frac{\mathbf{A}}{\|\mathbf{A}\|} = \mathbf{w}_{12d} \times \mathbf{b}_{3d}. \quad (16)$$

$$\mathbf{w}_{12d} = \mathbf{b}_{3d} \times \dot{\mathbf{b}}_{3d}. \quad (17)$$

The error variables $\mathbf{e}_b$ and $\mathbf{e}_w$ are defined as vectors in $\mathbb{R}^3$, whereas the integral terms e_{I1} and e_{I2} are defined as scalars in $\mathbb{R}$:

$$\mathbf{e}_b = \mathbf{b}_{3d} \times \mathbf{b}_3. \quad (18)$$

$$\mathbf{e}_w = \mathbf{w}_{12} + \mathbf{S}(\mathbf{b}_3)^2 \mathbf{w}_{12d}. \quad (19)$$

$$e_{I1} = \int_0^t (\mathbf{e}_w + m_1 \mathbf{e}_b) \mathbf{b}_1 dt \quad (20)$$

$$e_{I2} = \int_0^t (\mathbf{e}_w + m_1 \mathbf{e}_b) \mathbf{b}_2 dt \quad (21)$$

By differentiating the expression (19), we obtain:

$$\begin{aligned} \dot{\mathbf{e}}_w &= \dot{\mathbf{w}}_{12} + \mathbf{S}(\mathbf{b}_3)^2 \mathbf{w}_{12d} \\ &\quad + \dot{\mathbf{b}}_3(\mathbf{b}_3 \cdot \mathbf{w}_{12d}) + \mathbf{b}_3(\dot{\mathbf{b}}_3 \cdot \mathbf{w}_{12d}). \end{aligned} \quad (22)$$

To incorporate $\dot{\mathbf{w}}_{12}$ as defined in (15) into (22), we proceed as follows:

$$\begin{aligned} \dot{\mathbf{e}}_w &= (\dot{w}_{10} + D_1(w, \Delta t) + \frac{\Delta\tau_1}{J_1} + \frac{\Delta_1}{J_1}) \mathbf{b}_1 \\ &\quad + (\dot{w}_{20} + D_2(w, \Delta t) + \frac{\Delta\tau_2}{J_2} + \frac{\Delta_2}{J_2}) \mathbf{b}_2 \\ &\quad + \mathbf{S}(\mathbf{b}_3)^2 \mathbf{w}_{12d} + \dot{\mathbf{b}}_3(\mathbf{b}_3 \cdot \mathbf{w}_{12d}) + \mathbf{b}_3(\dot{\mathbf{b}}_3 \cdot \mathbf{w}_{12d}). \end{aligned} \quad (23)$$

Calculate the incremental control input torque as follows:

$$\begin{aligned} \frac{\Delta\tau_1}{J_1} \mathbf{b}_1 + \frac{\Delta\tau_2}{J_2} \mathbf{b}_2 &= -(\dot{w}_{10} + D_1(w, \Delta t)) \mathbf{b}_1 \\ &\quad - (\dot{w}_{20} + D_2(w, \Delta t)) \mathbf{b}_2 \\ &\quad - k_b \mathbf{e}_b - k_w \mathbf{e}_w - k_I e_{I1} \mathbf{b}_1 - k_I e_{I2} \mathbf{b}_2 \\ &\quad - (\mathbf{b}_3 \mathbf{w}_{12d}) \dot{\mathbf{b}}_{3d} - \mathbf{S}(\mathbf{b}_3)^2 \dot{\mathbf{w}}_{12d}. \end{aligned} \quad (24)$$

$$\dot{\mathbf{e}}_w = -k_b \mathbf{e}_b - k_w \mathbf{e}_w - \mathbf{F} + \mathbf{b}_3(\dot{\mathbf{b}}_3 \cdot \mathbf{w}_{12d}). \quad (25)$$

Where $\mathbf{F} = \sum_{j=1}^2 (k_I e_{Ij} - \frac{\Delta\tau_j}{J_j}) \mathbf{b}_j \in \mathbb{R}^3$.

Select the Lyapunov function V_1 in the following manner:

$$V_1 = \frac{1}{2} \mathbf{e}_w \cdot \mathbf{e}_w + m_1 \mathbf{e}_b \cdot \mathbf{e}_w + k_b(1 - \mathbf{b}_3 \cdot \mathbf{b}_{3d}) + \mathbf{F}. \quad (26)$$

By differentiating V_1 and simplifying the expression, we obtain:

$$\begin{aligned} \dot{V}_1 \leq & (1 - k_w) \|\mathbf{e}_w\|^2 + (B_2 + k_w m_1) \|\mathbf{e}_b\| \|\mathbf{e}_w\| \\ & + (-k_b m_1) \|\mathbf{e}_b\|^2. \end{aligned} \quad (27)$$

Define $\mathbf{z}_1 = (\|\mathbf{e}_b\|, \|\mathbf{e}_w\|)^T \in \mathbb{R}^2$, then (27) can be simplified to:

$$\dot{V}_1 \leq \mathbf{z}_1^T \mathbf{Q}_1 \mathbf{z}_1. \quad (28)$$

where the following represents the matrix $\mathbf{Q}_1 \in \mathbb{R}^{2 \times 2}$:

$$\mathbf{Q}_1 = \begin{bmatrix} -k_b m_1 & \frac{1}{2}(B_2 - k_w m_1) \\ \frac{1}{2}(B_2 - k_w m_1) & 1 - k_w \end{bmatrix}. \quad (29)$$

3.2 Yaw Control Subsystem

This article assumes that $\mathbf{b}_{1d}(t)$ represents the desired direction of the first body axis of a quadrotor UAV, as explained in Section 3.1. The normal plane of the $\mathbf{b}_3$ axis is projected with the desired direction of the first body axis $\mathbf{b}_{1d}$:

$$\begin{aligned} \mathbf{b}_{1c}(t) &= -\mathbf{S}(\mathbf{b}_3)^2 \mathbf{b}_{1d}(t) \\ &= (I_{3 \times 3} - \mathbf{b}_3(t) \mathbf{b}_3(t)^T) \mathbf{b}_{1d}(t). \end{aligned} \quad (30)$$

Therefore, $\mathbf{b}_{1c} \cdot \mathbf{b}_3 = 0$. The objective of yaw control is to design τ_3 such that as $t \rightarrow \infty$, $\mathbf{b}_1 \rightarrow \mathbf{b}_{1c}$.

The following methods allow us to acquire $\dot{\mathbf{b}}_1$ and $\dot{\mathbf{b}}_{1c}$ similarly to (10):

$$\dot{\mathbf{b}}_1 = \mathbf{w}_{23} \times \mathbf{b}_1 = -\mathbf{w}_2 \mathbf{b}_3 + \mathbf{w}_3 \mathbf{b}_2. \quad (31)$$

$$\begin{aligned} \dot{\mathbf{b}}_{1c} &= \mathbf{w}_c \times \mathbf{b}_{1c} \\ &= (w_{c1} \mathbf{b}_1 + w_{c2} \mathbf{b}_2 + w_{c3} \mathbf{b}_3) \times \mathbf{b}_{1c}. \end{aligned} \quad (32)$$

Define the following error function for the UAV's yaw subsystem:

$$\Omega(\mathbf{b}_1, \mathbf{b}_{1c}) = \frac{1}{2} \|\mathbf{b}_1 - \mathbf{b}_{1c}\|^2 = 1 - \mathbf{b}_1 \cdot \mathbf{b}_{1c}. \quad (33)$$

By using the derivative of the UAV's yaw subsystem's error function (33) we obtain:

$$\dot{\Omega}(\mathbf{b}_1, \mathbf{b}_{1c}) = (\mathbf{w}_3 - \mathbf{w}_{c3})(-\mathbf{b}_2 \cdot \mathbf{b}_{1c}). \quad (34)$$

This leads to the following fault definition for the UAV's yaw subsystem:

$$e_\Psi = -\mathbf{b}_2 \cdot \mathbf{b}_{1c}. \quad (35)$$

$$e_{w\Psi} = \mathbf{w}_3 - \mathbf{w}_{c3}. \quad (36)$$

The simplified equation is as follows after expanding (8) in a Taylor series:

$$\dot{w}_3 = \dot{w}_{30} + \frac{\Delta\tau_3}{J_3} + \frac{\Delta_3}{J_3}. \quad (37)$$

By taking the derivative of (36) and replacing it with (37), we obtain:

$$\dot{e}_{w\Psi} = \dot{w}_{30} + \frac{\Delta\tau_3}{J_3} + \frac{\Delta_3}{J_3} - \dot{w}_{c3}. \quad (38)$$

As a result, the UAV's yaw subsystem's intended incremental control input torque is:

$$\frac{\Delta\tau_3}{J_3} = -\dot{w}_{30} + \dot{w}_{c3} - k_\Psi e_\Psi - k_{w\Psi} e_{w\Psi}. \quad (39)$$

Substituting (39) into (38), we get:

$$\dot{e}_{w\Psi} = -k_\Psi e_\Psi - k_{w\Psi} e_{w\Psi} + \frac{\Delta_3}{J_3}. \quad (40)$$

Select the V_2 Lyapunov function:

$$V_2 = \frac{1}{2} \mathbf{e}_{w\Psi} \cdot \mathbf{e}_{w\Psi} + \mathbf{e}_\Psi \cdot \mathbf{e}_{w\Psi} + k_3(1 - \mathbf{b}_1 \cdot \mathbf{b}_{1c}). \quad (41)$$

After applying (41) to the derivative of the Lyapunov function V_2 , we obtain:

$$\begin{aligned} \dot{V}_2 &\leq (1 - k_{w\Psi}) \|\mathbf{e}_{w\Psi}\|^2 - k_{w\Psi} \|\mathbf{e}_{w\Psi}\| \|\mathbf{e}_\Psi\| \\ &\quad - k_3 \|\mathbf{e}_\Psi\|^2 + (\|\mathbf{e}_\Psi\| + \|\mathbf{e}_{w\Psi}\|) \frac{\Delta_3}{J_3}. \end{aligned} \quad (42)$$

Define $\mathbf{z}_2 = (\|\mathbf{e}_\Psi\|, \|\mathbf{e}_{w\Psi}\|)^T \in \mathbb{R}^2$, then we can see that $\|\mathbf{e}_\Psi\| + \|\mathbf{e}_{w\Psi}\| \leq \sqrt{2} \|\mathbf{z}_2\|$. Therefore, (42) can be simplified as:

$$\begin{aligned} \dot{V}_2 &\leq \mathbf{z}_2^T \mathbf{Q}_2 \mathbf{z}_2 + \alpha \|\mathbf{z}_2\| \\ &\leq \lambda_{\min}(\mathbf{Q}_2) \|\mathbf{z}_2\|^2 + \alpha \|\mathbf{z}_2\| \\ &\leq \rho \|\mathbf{z}_2\|^2, \|\mathbf{z}_2\| \geq \frac{\alpha}{\lambda_{\min}(\mathbf{Q}_2) - \rho}. \end{aligned} \quad (43)$$

where $\rho \in (\lambda_{\min}(\mathbf{Q}_2), 0)$, and the matrix $\mathbf{Q}_2 \in \mathbb{R}^{2 \times 2}$ is given by:

$$\mathbf{Q}_2 = \begin{bmatrix} -k_3 c_2 & -\frac{1}{2} k_{w\Psi} \\ -\frac{1}{2} k_{w\Psi} & 1 - k_{w\Psi} \end{bmatrix}. \quad (44)$$

4 Simulation

To assess the efficacy of the decoupled geometric control algorithm utilizing INDI, a comparative simulation is conducted in MATLAB, comparing the proposed method with an uncoupled geometric control algorithm. The quadrotor

UAV simulation is provided with an intended trajectory, denoted as (45):

$$\begin{cases} w_{ref} = 2\pi/T(\text{rad/s}) \\ \mathbf{x}_d(t) = [A \sin(a t + d), B \sin(b t), 1 + C \cos(c t)]^T \\ \Psi_d(t) = [\cos(w_{ref} t), \sin(w_{ref} t), 0]^T \end{cases} \quad (45)$$

Figure 1 shows the trajectory of the quadrotor UAV in the MATLAB simulation, represented in three dimensions. In the diagram, $\mathbf{p}_d$ denotes the desired trajectory as defined in (45), $\mathbf{p}_1$ represents the three-dimensional trajectory computed by the algorithm presented in this paper, and $\mathbf{p}_2$ represents the trajectory computed using the geometric control algorithm defined in [13]. Figures 2-3 present the numerical simulation results corresponding to this information.

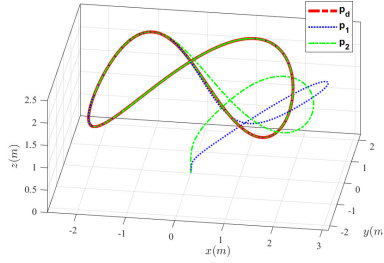


Fig. 1: The 3D trajectory of the quadrotor UAV

Figure 2 illustrates the positional tracking error of the quadrotor UAV in the MATLAB simulation. The subscript 1 indicates the tracking error achieved by the algorithm described in this paper, while the subscript 2 indicates the tracking error achieved by the comparative algorithm. The algorithm presented in this paper demonstrates superior performance in terms of convergence speed and tracking accuracy, outperforming the comparator algorithm in position tracking.

Figure 3 depicts the errors in the attitude angle. The subscript 1 indicates the attitude angle error derived from the algorithm described in this paper, while the subscript 2 reflects the attitude angle error obtained from the comparative algorithm. The simulation findings indicate that the algorithm proposed in this paper outperforms the comparative algorithm in terms of tracking speed, tracking performance, and tracking error magnitude.

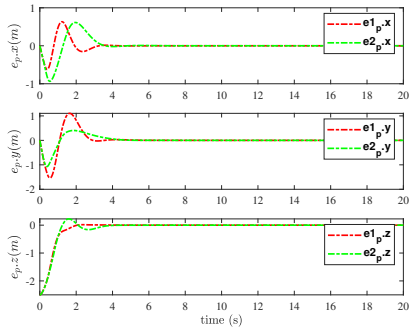


Fig. 2: The position tracking errors of the quadrotor UAV

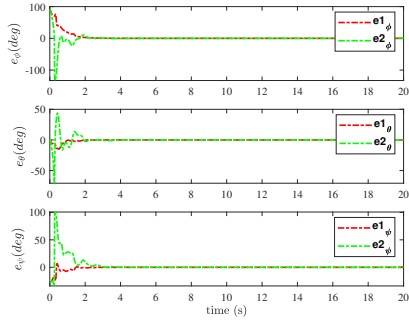


Fig. 3: The attitude tracking errors of the quadrotor UAV

5 Conclusion

This paper introduces a decoupled geometric control algorithm based on INDI. The method allows the quadrotor UAV to achieve agile flight even when facing unknown external disturbances. In contrast to current non-decoupled geometric control algorithms, the proposed algorithm effectively linearizes the UAV's nonlinear model by employing INDI. This results in enhanced response time and improved accuracy in tracking the trajectory. Future work will focus on applying the proposed control algorithm to practical quadrotor UAVs.

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